

IMPROVING SKIN LESION IMAGE CLASSIFICATION USING PATIENT DATA

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Early Thoughts

Context/Motivation

Skin cancer detection is an important medical imaging problem because early identification of malignant lesions, especially melanoma, can significantly affect patient outcomes. Dermatologists often rely on both visual inspection of lesions and contextual clinical information, such as patient age and lesion location, when making diagnostic judgments. Machine learning would be great for processing visuals and adding these contextual elements, and perhaps we can explore the importance of this in the medical field.

Research Question

Can dermoscopic skin lesion images be used to accurately classify lesions as benign or malignant, and does the inclusion of metadata such as age, sex, and lesion location improve classification performance relative to an image-only approach?

Hypothesis

Dermoscopic skin lesion images will allow a model to classify lesions as benign or malignant with meaningful predictive accuracy, but that an image-plus-metadata model will outperform an image-only model. Adding patient metadata (age, sex, and lesion location) will improve performance on clinically important metrics, especially recall for malignant lesions, because diagnostic decisions in practice rely not only on visual patterns in the lesion image but also on contextual clinical information.

Data Acquisition

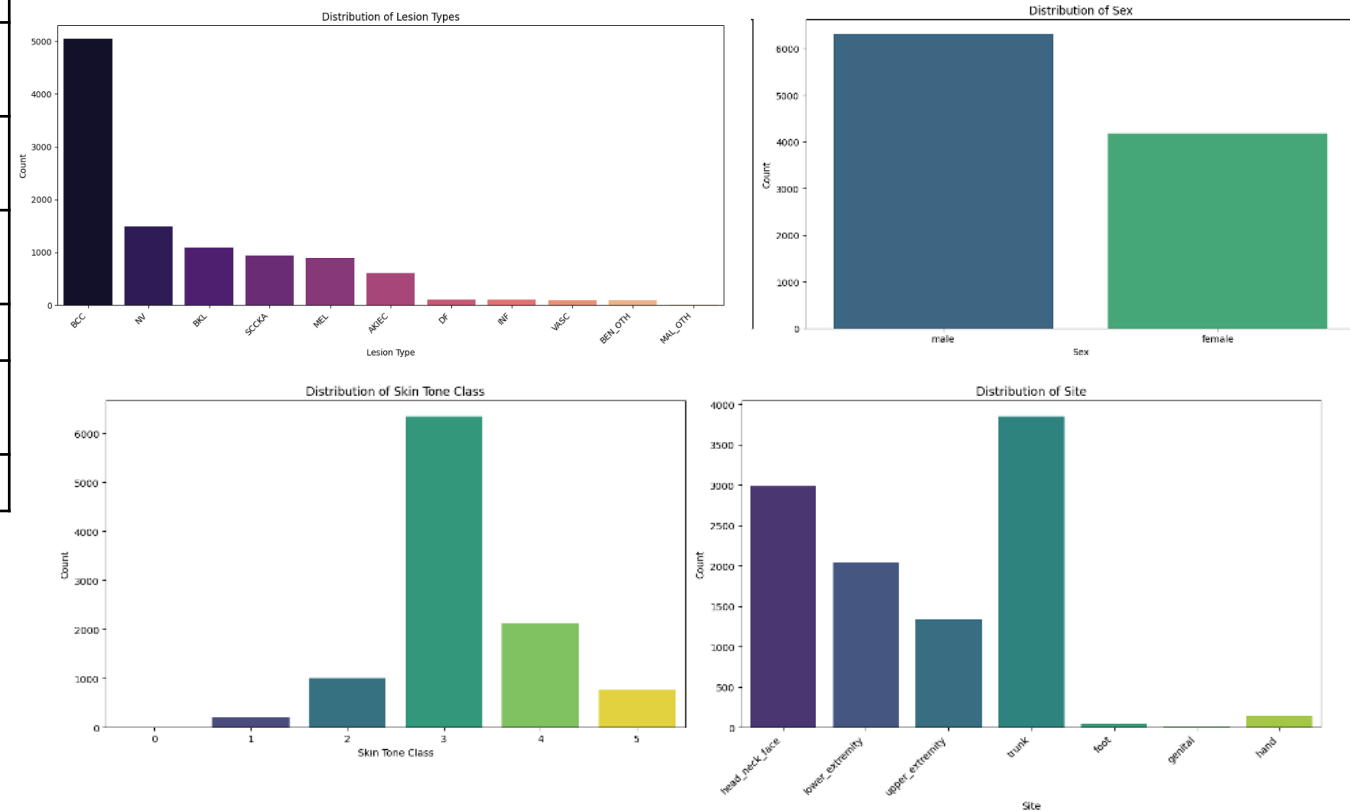
Column name	Description	Example
lesion_id	Unique identifier for each skin lesion	IL_0000652
image_type	Type of image captured (clinical photo or dermoscopic image)	clinical: close-up
isic_id	Unique image identifier from the ISIC database	ISIC_8149219
attribution	Source or contributing group responsible for the data	MILK study team
copyright_license	License governing the use of the image/data	CC-BY-NC
image_manipulation	Indicates whether the image was altered or instrument-derived	altered
age_approx	Approximate age of the patient at time of image capture (5-year intervals)	70
sex	Biological sex of the patient	male
skin_tone_class	Numeric classification of patient skin tone (e.g., Fitzpatrick scale grouping)	1
site	Body location where the lesion is found	head_neck_face

Data Source:

International Skin Imaging Collaboration (ISIC)

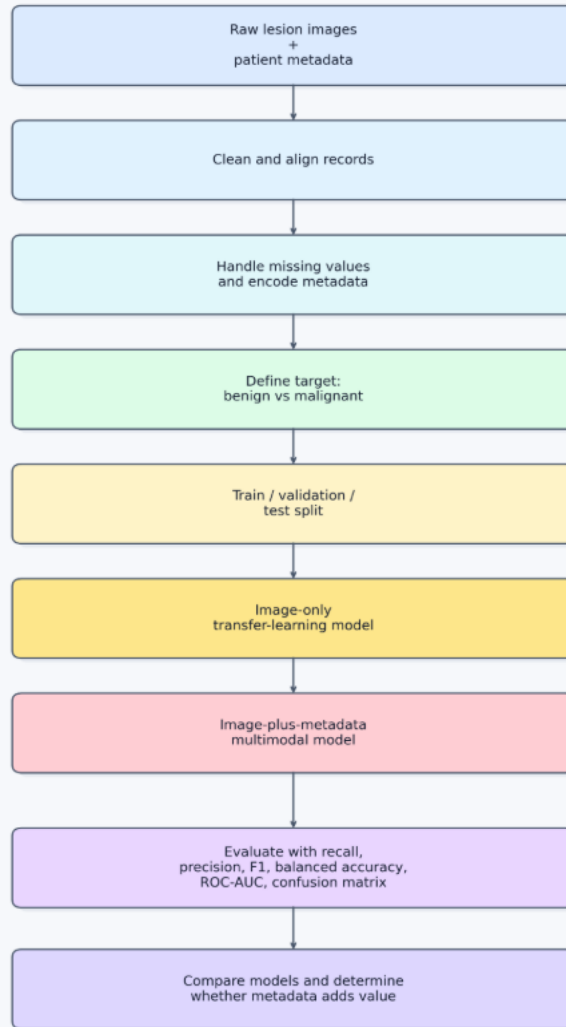
Preprocessing:

- **Image Matching:** Used isic_id to pair clinical and dermoscopic views for every case.
- **Patient-Level Grouping:** Linked images to metadata via lesion_id to ensure consistent patient tracking.
- **Data Integrity:** Used lesion_id for stratified splitting to prevent training/testing data leakage.
- **Metadata Selection:** Focused on approx_age, sex, and site as the most reliable clinical predictors.



Analysis Plan

Graphical Representation of Analysis Plan



Model Features

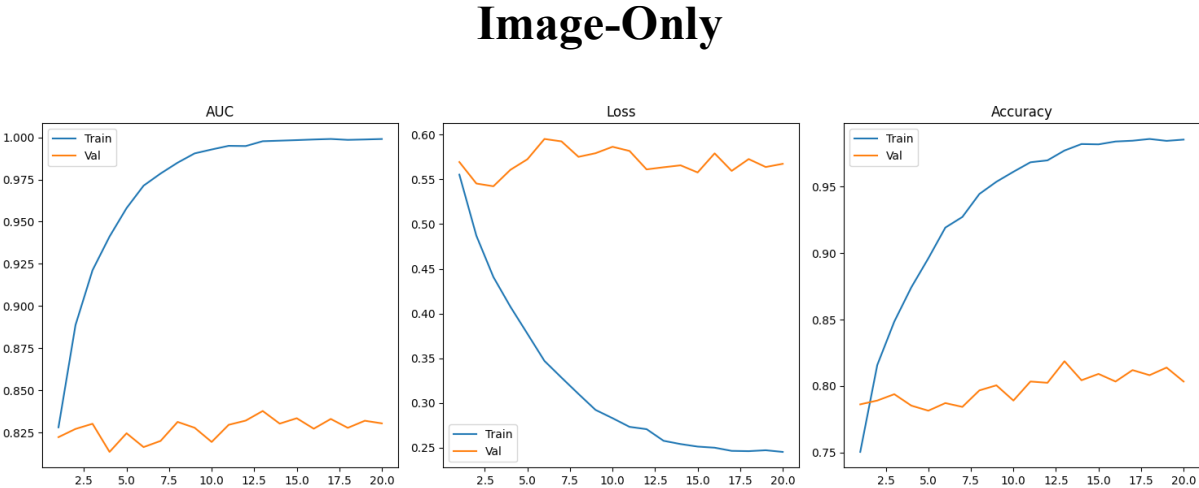
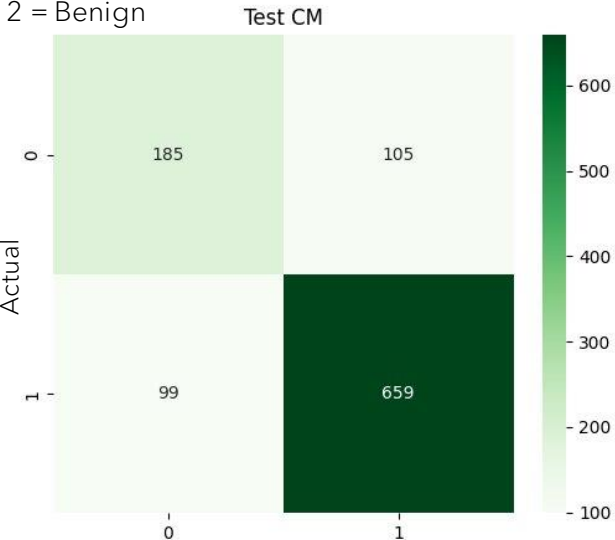
- **EfficientNet-B0 (timm):** Highly pre-trained backbone used for efficient, high-performance visual feature extraction.
- **Late Fusion MLP:** A network layer that merges metadata with image features for final classification.
- **Stratified Lesion-Level Split:** Grouping images by patient ID to prevent data leakage and ensure fair testing.
- **Weighted Cross-Entropy:** A customized loss function designed to prioritize rare and dangerous malignant classes.
- **ImageNet Normalization:** Standardizes input pixels to match the pre-trained model's expected color scales.

Tricky Decisions

- **Metric Selection:** We chose to save the best model based on AUC rather than accuracy to prevent the lopsided dataset from inflating our performance scores.
- **Class Imbalance:** We prioritized specific weighting for rare malignant cases to ensure the model didn't default to predicting Basal Cell Carcinoma, the most common class.
- **Image Integration:** We chose to combine clinical and dermoscopic images into a single input because this paired approach outperformed analyzing the images separately.

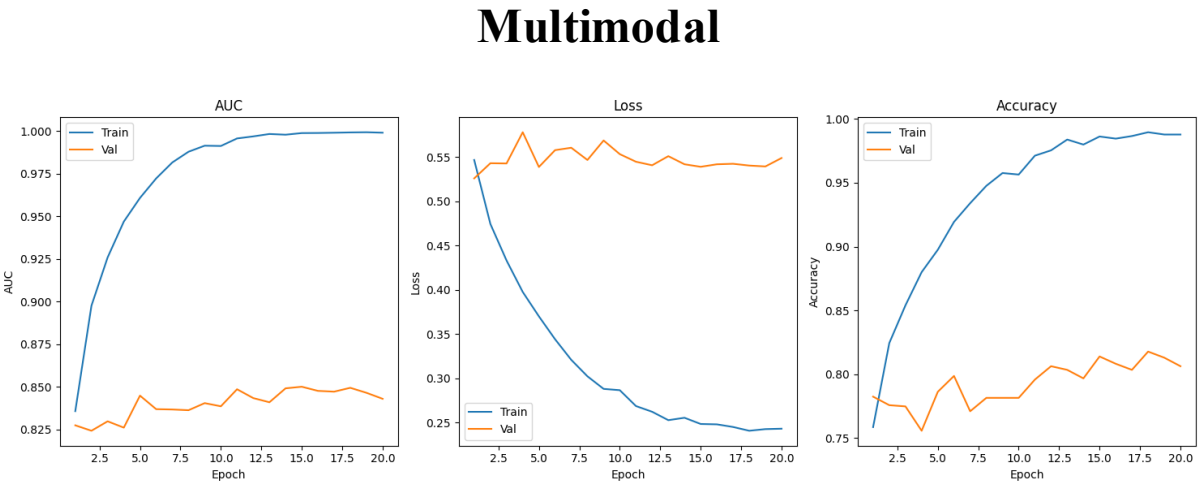
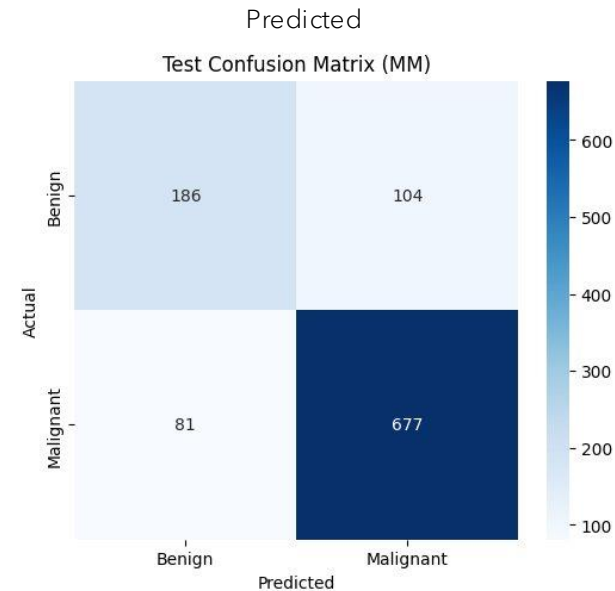
Binary Results: Metadata Drives 2% Boost in Malignant Recall

1 = Malignant
2 = Benign



--- Final Evaluation (Image-Only) ---
Test AUC: 0.8247 | Test Acc: 0.8053

	precision	recall	f1-score	support
Benign	0.65	0.64	0.64	290
Malignant	0.86	0.87	0.87	758
accuracy			0.81	1048
macro avg	0.76	0.75	0.76	1048
weighted avg	0.80	0.81	0.80	1048

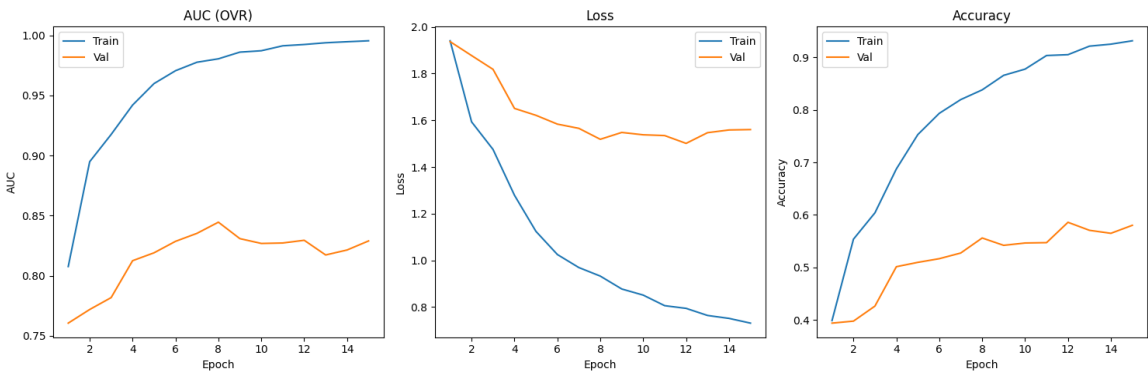
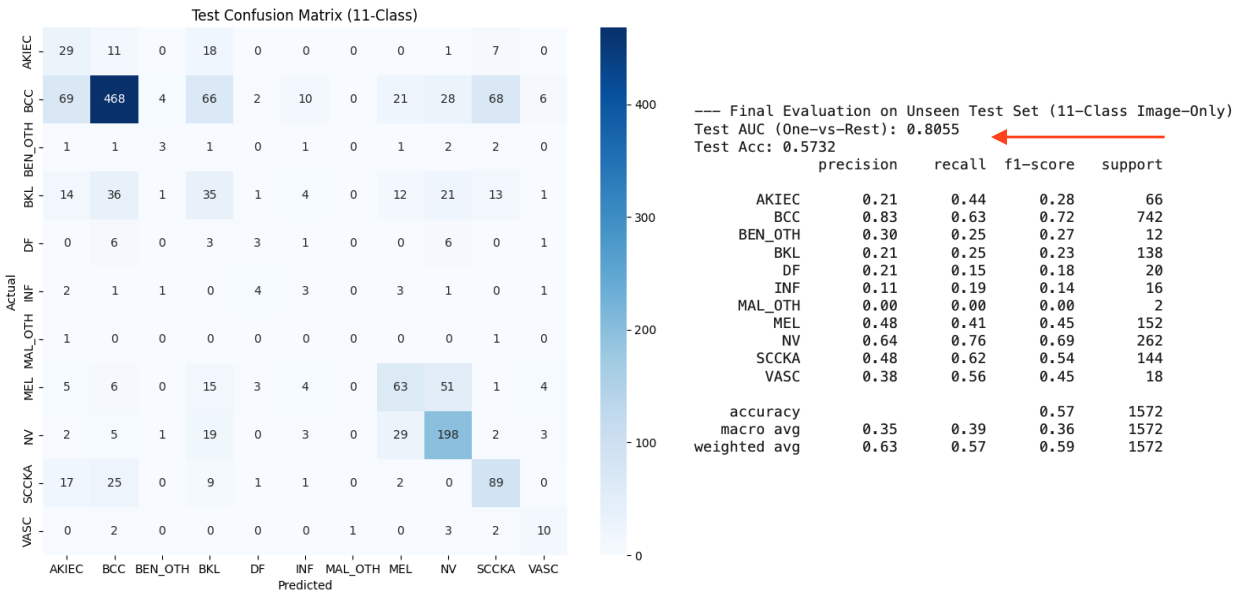


--- Final Evaluation (Multimodal) ---
Test AUC: 0.8489 | Test Acc: 0.8235

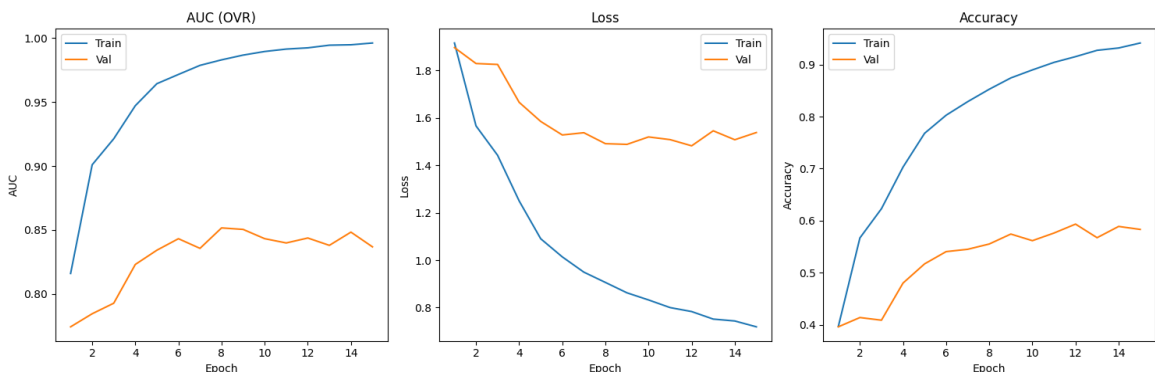
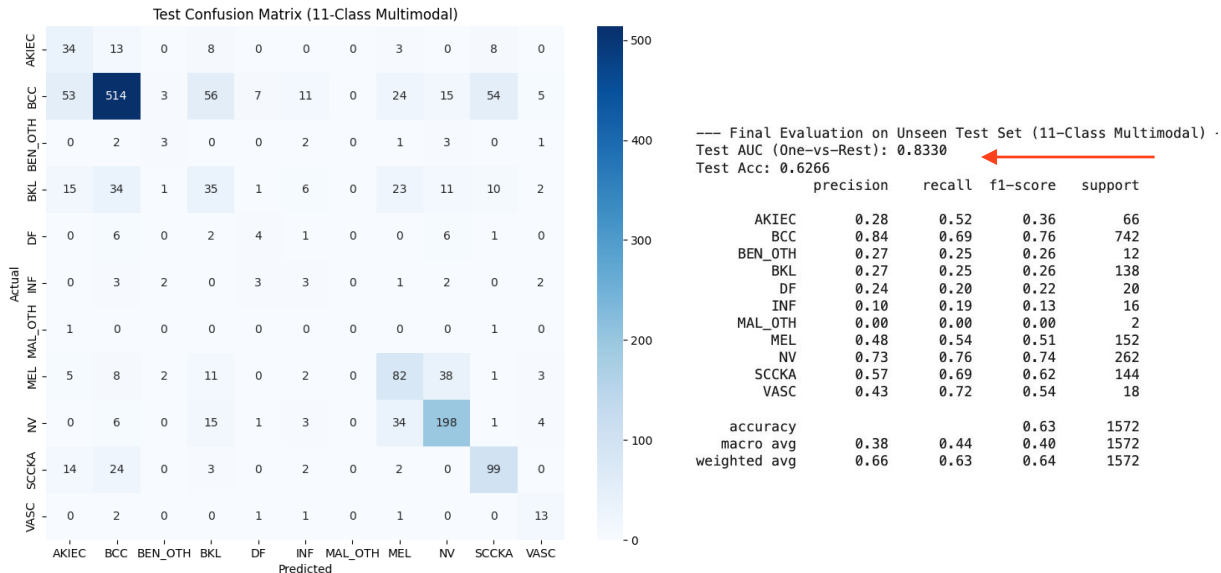
	precision	recall	f1-score	support
Benign	0.70	0.64	0.67	290
Malignant	0.87	0.89	0.88	758
accuracy			0.82	1048
macro avg	0.78	0.77	0.77	1048
weighted avg	0.82	0.82	0.82	1048

11-Class Results: Significant 5.34% Accuracy Gain via Metadata

Image-Only



Multimodal



Statistical Significance & Results

Performance was measured using **Accuracy** and **One-vs-Rest AUC**. Comparison between the Image-Only and Multimodal architectures was conducted using:

- **Two-Proportion Z-Test:** To evaluate the significance of Accuracy deltas.
- **Hanley & McNeil Method:** To evaluate the significance of AUC shifts.
- **Significance Threshold:** $\alpha = 0.05$

Experiment	Metric	Image-Only	Multimodal	Difference	P-Value	Significance
Binary	Accuracy	80.53%	82.35%	+1.82%	0.2839	Not Significant
Binary	AUC	0.8247	0.8489	+0.0242	0.1337	Not Significant
11-Class	Accuracy	57.32%	62.66%	+5.34%	0.0022	Highly Significant
11-Class	AUC	0.8055	0.8330	+0.0275	0.0450	Significant

- Metadata acts as a diagnostic tie-breaker because visual features often overlap between distinct pathologies, which explains the highly significant 5.34% accuracy boost in the 11-class model.
- The lack of statistical significance in the binary model suggests that visual pixels are sufficient for general screening, but demographic context is required for specific disease differentiation.
- The highest performance gains were concentrated in categories where visual ambiguity is most common, specifically for vascular lesions and squamous cell carcinoma.
 - The primary clinical success of the multimodal approach was the reduction of false negatives in malignant classes, making the system a safer tool for high-stakes decision-making.



Limitations & Biases

- **Class imbalance:** Some lesion categories are much more common than others, which can make a model look stronger overall than it really is on rarer malignant cases.
- **Potential overfitting:** Medical image models can fit too closely to dataset-specific patterns, especially with transfer learning and limited rare-class examples.
- **Dataset bias:** The data may not fully represent all patient groups, skin tones, or clinical settings, which can limit generalizability.
- **Artifact risk:** Some image-derived features may reflect markings, lighting, or dermoscopy artifacts rather than actual lesion pathology.
- **Binary simplification:** Focusing mainly on benign vs. malignant improves clarity, but it may hide differences across specific lesion subtypes.

Next Steps

- **Improve Data Diversity:** Collect more images from different skin tones and clinical settings to ensure the model works fairly and accurately for all patients.
- **Focus on Rare Diseases:** Develop better ways for the model to identify rare but dangerous cancers that are currently outnumbered by common, harmless spots.
- **Prevent Over-Learning:** Use better training techniques to make sure the model is learning actual medical signs rather than just memorizing specific patterns in this dataset.
 - **Identify Key Factors:** Figure out which specific pieces of patient information (like age or location) are most helpful for the model's decision-making.
- **Test in Real Settings:** Move toward testing the model as a helpful assistant for doctors to see how it performs in a real-world medical environment.

References

- [1] Tschandl P., Rosendahl C. & Kittler H. The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. Sci. Data 5, 180161 doi.10.1038/sdata.2018.161 (2018)
- [2] Noel C. F. Codella, David Gutman, M. Emre Celebi, Brian Helba, Michael A. Marchetti, Stephen W. Dusza, Aadi Kalloo, Konstantinos Liopyris, Nabin Mishra, Harald Kittler, Allan Halpern: "Skin Lesion Analysis Toward Melanoma Detection: A Challenge at the 2017 International Symposium on Biomedical Imaging (ISBI), Hosted by the International Skin Imaging Collaboration (ISIC)", 2017; arXiv:1710.05006.
- [3] Hernández-Pérez C, Combalia M, Podlipnik S, Codella NC, Rotemberg V, Halpern AC, Reiter O, Carrera C, Barreiro A, Helba B, Puig S, Vilaplana V, Malvehy J. BCN20000: Dermoscopic lesions in the wild. Scientific Data. 2024 Jun 17;11(1):641.

ANY QUESTIONS?

